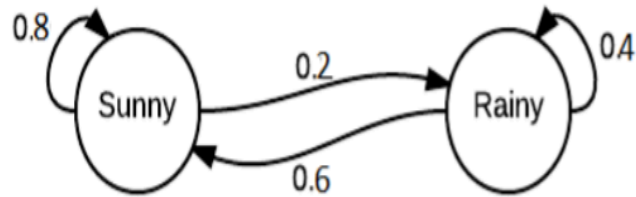


1, Given a weather model

Try to calculate the percentage of each weather and fill the following table.



For All data			
	Total	Sunny	Rainy
Percentage	100%	75%	25%

What is the weather after a Rainy day			
	Total	Sunny	Rainy
Percentage	100%	60%	40%

What is the weather before a sunny day			
	Total	Sunny	Rainy
Percentage	100%	80%	20%

2, Here is the Hidden Markov model for the Dishonest Casino problem.

A casino has two dice:

► Fair die

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$$

► Loaded die

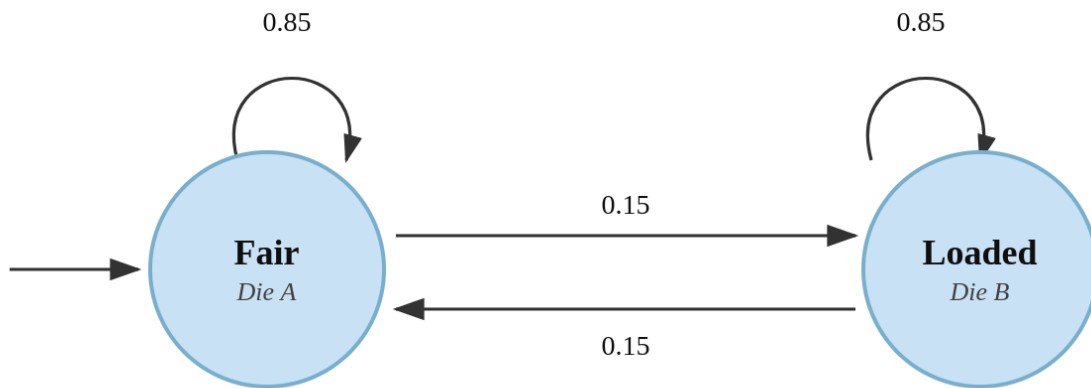
$$P(1) = P(2) = P(3) = P(4) = P(5) = 1/15$$

$$P(6) = 2/3$$

The Transition probability (switching between fair and loaded die) is 0.15

Note, use base 2 for all the log calculation.

2.1, Draw the FSA (finite state automata) for this Hidden Markov model.



$P(\text{face} \mid \text{Fair})$		
1 : $\frac{1}{6}$	2 : $\frac{1}{6}$	3 : $\frac{1}{6}$
4 : $\frac{1}{6}$	5 : $\frac{1}{6}$	6 : $\frac{1}{6}$

$P(\text{face} \mid \text{Loaded})$		
1 : $\frac{1}{15}$	2 : $\frac{1}{15}$	3 : $\frac{1}{15}$
4 : $\frac{1}{15}$	5 : $\frac{1}{15}$	6 : $\frac{2}{3}$

2.2, Given the HMM, and a sequence of rolls by the casino as following:

6, 2, 4, 6, 2, 3, 6, 1, 2, 1, 5, 6, 2, 1, 4

What is the probability of states “Fair, Fair, Fair, Fair, Fair, Load, Load, Load, Load, Fair, Fair, Fair, Fair, Fair, Fair”

The probability of the given sequence (FFFFFLLLLFFFFF) = $8.713 \times 10^{(-16)}$

(Using Programs we have already made)

2.3, Given the HMM, and a sequence of rolls by the casino “6, 2, 4, 6, 2, 3, 6, 1, 2, 1, 5, 6, 2, 1, 4”, try to predict the state of each roll. (generated by the loaded/fair die)

The predicted state of each roll is: FFFFFFFFFFFFFFFFFF

(Using Programs we have already made)

3. “Man in the Iron Mask” described a story that the identical brother of Louis XIV of France, man in the iron mask, was liberated by former musketeer and then installed onto the throne.

Can you design a HMM to figure out their identities based on the behavior patterns? How?

This question is perfect for an HMM. The hidden states, similar to the dishonest casino, would be either of the twins. There would be probabilities of the twins swapping out for each other but it would likely be low such as 0.01 since rulers wouldn't change that often. The emission states of this model could be behaviors that either twin exhibits, such as speech patterns, rulings, where the king (twin) is spotted, etc.

To figure out which twin is ruling we can take observed behaviors that I mentioned above, and use the Viterbi algorithm to find the most likely sequence of which twin was on the throne at each point in time.

The forward-backward algorithm could also compute the probability $P(\text{Twin} \mid \text{observations})$ at each point in time to determine if and when a swap occurred.